THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 13: **Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The chemicals that caused the ozone hole are: 1

(a) chlorofluorocarbons (CFCs) **(b)** nitrogen
(c) carbon dioxide **(d)** methane

2. The largest reservoir of nitrogen is the: 1

(a) oceans **(b)** plants **(c)** atmosphere **(d)** soil

3. The envelope of gases around the Earth is the: 1

(a) atmosphere **(b)** hydrosphere **(c)** biosphere **(d)** geosphere

4. The equator receives more heat than the poles because: 1

(a) it has less atmosphere **(b)** it has more ice
(c) the Sun's rays fall almost vertically **(d)** it is farther from the Sun

5. The international agreement to phase out ozone-depleting substances is the: 1

(a) Montreal Protocol **(b)** Geneva Convention
(c) Paris Agreement **(d)** Kyoto Protocol

6. The bacteria that convert atmospheric nitrogen into usable forms are: 1

(a) Lactobacillus **(b)** nitrogen-fixing bacteria
(c) decomposers only **(d)** E. coli

7. All living organisms on the Earth together with their environment form the: 1

(a) atmosphere **(b)** biosphere **(c)** geosphere **(d)** hydrosphere

8. The gases that trap heat in the atmosphere are called: 1

(a) fuel gases **(b)** inert gases **(c)** greenhouse gases **(d)** noble gases

9. The ozone hole was first observed over: 1

(a) Antarctica **(b)** the Arctic only **(c)** Africa **(d)** India

10. The process by which plants release water vapour is: 1

(a) transpiration **(b)** respiration **(c)** condensation **(d)** photosynthesis

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Name two ways in which an individual can reduce greenhouse gas emissions. 2

12. What is albedo? Give an example of a high-albedo surface. 2

13. Why do polar regions receive less heat than the equator? 2

14. Why is UV radiation both harmful and useful? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. State the causes and three effects of ozone depletion. 3

16. State three effects of global warming. 3

17. Describe how humans have changed the natural cycles of carbon and nitrogen. 3

OR

Describe the carbon cycle in brief. 3

18. What are the CFCs? State their uses and the steps taken to reduce them. 3

OR

Explain why the speed of light matters in energy: calculate the time light takes to travel 1.5 x 10^11 m at 3 x 10^8 m/s. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the greenhouse effect, the greenhouse gases, the sources of each and the consequences of global warming, with the mitigation measures. 5

OR

Explain the potential of solar energy in India and the devices that use it, with the advantages and limitations. 5

20. Explain the ozone layer, its depletion, the effects and the international response. 5

OR

Explain the importance of Mission LiFE and describe the actions that students can take in their school and community. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND